

Lecture 09

Diagonalization

Consider the matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 0 & 3 \end{bmatrix}.$$

What is the characteristic polynomial of A ?

- ✓ 1. $(1 - \lambda)(2 - \lambda)(3 - \lambda)$
2. $(1 - \lambda)(2 - \lambda)(3 - \lambda) - (1)(1)(1)(2)(4)(3)$

Recall: $\det \begin{bmatrix} a & * & * \\ 0 & b & * \\ 0 & 0 & c \end{bmatrix} = a \cdot b \cdot c$

char poly

is $\det(\hat{A} - \lambda \hat{I})$

$$= \det \begin{bmatrix} 1-\lambda & 1 & 1 \\ 0 & 2-\lambda & 4 \\ 0 & 0 & 3-\lambda \end{bmatrix}$$

$$= (1-\lambda)(2-\lambda)(3-\lambda)$$

$$\lambda = 1$$

$$\lambda = 2$$

$$\lambda = 3$$

Example. Consider the matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 0 & 3 \end{bmatrix}.$$

What are its eigenvalues and its corresponding eigenvectors?

(Step 1) Find eigenvalues by finding all roots of its **characteristic polynomial**

$$\det(A - \lambda I).$$

(Step 2) For each eigenvalue λ , determine the corresponding **eigenspace**.

$$E_\lambda = \text{null}(A - \lambda I).$$

Any nonzero vector in E_λ is an eigenvector corresponding to λ .

$$(\lambda = 1) \quad E_1 = \text{null}(A - 1 \cdot I)$$

$$\text{Solve } \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 4 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \begin{matrix} \text{ref} \\ \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$\text{So, we have } E_1 = \left\{ s \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} : s \in \mathbb{R} \right\}$$

$\therefore \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ is an eigenvector with eigenvalue 1.

DIY

$$\lambda = 2 \quad E_2 = \left\{ s \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} : s \in \mathbb{R} \right\}$$

$$\lambda = 3 \quad E_3 = \left\{ s \begin{bmatrix} 5 \\ 8 \\ 2 \end{bmatrix} : s \in \mathbb{R} \right\}$$

Définition. Let A be a real $n \times n$ -matrix.

We say that A is *diagonalizable* if there exists

an invertible matrix P and a diagonal matrix D such that

$$A = PDP^{-1}.$$

Example. Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 0 & 3 \end{bmatrix}$. Recall that its characteristic polynomial is $(1 - \lambda)(2 - \lambda)(3 - \lambda)$. Recall

$$\bullet E_1 = \left\{ s \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} : s \in \mathbb{R} \right\}$$

$$\bullet E_2 = \left\{ s \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} : s \in \mathbb{R} \right\}$$

$$\bullet E_3 = \left\{ s \begin{bmatrix} 5 \\ 8 \\ 2 \end{bmatrix} : s \in \mathbb{R} \right\}$$

$$\hat{P} = \begin{bmatrix} 1 & 1 & 5 \\ 0 & 1 & 8 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\hat{D} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Check:

$$\hat{A} = \hat{P} \hat{D} \hat{P}^{-1}$$

Equivalently

$$\hat{A} \hat{P} = \hat{P} \hat{D}$$

[DIY
see Lec 8A]

Recall: TFAE $\rightarrow$ the following are equivalent

- $\hat{P}$ is invertible
- cols of $\hat{P}$ are LoI
- cols of $\hat{P}$ form a basis for $\mathbb{R}^n$

Lecture 09

Diagonalization

- How to diagonalize?
- When can? When cannot?

Applications of Diagonalization

Eigen vectors

got some proof on colab

Diagonalization Theorem.

Let A be a real $n \times n$ -matrix. Then TFAE.

- (I) We have that A is diagonalizable.
- (II) There exists n eigenvectors of A that form a basis for $\mathbb{R}^n$.
- (III) For each eigenvalue λ_i of A , we have

$$\alpha(\lambda_i) = \dim(E_{\lambda_i}) \text{ and } \sum \dim(E_{\lambda_i}) = n.$$

Definition. Let A be a real $n \times n$ -matrix.

We say that A is *diagonalizable* if there exists an invertible matrix P and a diagonal matrix D such that

$$A = PDP^{-1}.$$

Def

$\alpha(\lambda_i)$ = algebraic multiplicity of λ_i

$\dim(E_{\lambda_i})$ = geometric multiplicity of λ_i .

Example. Consider the following matrix

→ dimension
 $n=3$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}.$$

Recall that its characteristic polynomial is $\lambda^2(14 - \lambda)$.

Notation.

$\alpha(\lambda_i)$ = algebraic multiplicity of λ_i

$\dim(E_{\lambda_i})$ = geometric multiplicity of λ_i

Two eigenvalues

$$\lambda_1 = 14, \alpha(\lambda_1) = 1$$

$$\lambda_2 = 0, \alpha(\lambda_2) = 2$$

(multiplicity
in the
usual sense)

↓
no. of times this root is repeated

Technical definition of
alg. multiplicity is too abstract,
just ignore

Example. Consider the following matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}.$$

Recall that its characteristic polynomial is $\lambda^2(14 - \lambda)$.

Recall

$$\lambda_1 = 14 \Rightarrow E_{14} = \left\{ s \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} : s \in \mathbb{R} \right\} \Rightarrow \dim(E_{14}) = 1$$

$$\lambda_2 = 0 \Rightarrow E_0 = \left\{ s \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} : s, t \in \mathbb{R} \right\} \Rightarrow \dim(E_0) = 2$$

Notation.

$\alpha(\lambda_i)$ = algebraic multiplicity of λ_i

$\dim(E_{\lambda_i})$ = geometric multiplicity of λ_i

2 vectors

$$\alpha(14) = 1 = \dim(E_{14})$$

$$\alpha(0) = 2 = \dim(E_0)$$

$$\text{Sum of multiplicity} = 3 = n$$

From Diag. Thm,
 $\hat{A}$ is diagonalisable.

Example. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$. Recall that its characteristic polynomial is $\lambda^2(14 - \lambda)$.

Recall

- $E_{14} = \left\{ s \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} : s \in \mathbb{R} \right\}$. So, $\dim(E_{14}) = 1 = \alpha(14)$.
- $E_0 = \left\{ s \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} : s, t \in \mathbb{R} \right\}$. So, $\dim(E_0) = 2 = \alpha(0)$.

Diagonalization Theorem.

Let A be a real $n \times n$ -matrix. Then TFAE.

- (I) We have that A is diagonalizable.
- (II) There exists n eigenvectors of A that form a basis for $\mathbb{R}^n$.
- (III) For each eigenvalue λ_i of A , we have $\alpha(\lambda_i) = \dim(E_{\lambda_i})$ and $\sum \dim(E_{\lambda_i}) = n$.

- Set $P = \begin{bmatrix} 1 & -2 & -3 \\ 2 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}$

- Set $D = \begin{bmatrix} 14 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

- So, $A = PDP^{-1}$ and A is diagonalizable.

Diagonalization

Let A be an $n \times n$ -matrix.

look at characteristic
poly nomial

(Step 1) Find all eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_s$.

→ algebraic multiplicities

- (Step 1a) If $\sum_{i=1}^k \alpha(\lambda_i) < n$, then A is **not** diagonalizable!

- (Step 1b) For each λ_i , check that $\alpha(\lambda_i) = \dim(E_{\lambda_i})$.

>>>> If $\alpha(\lambda_i) > \dim(E_{\lambda_i})$ for some i , then A is **not** diagonalizable!

>>>> If $\alpha(\lambda_i) = \dim(E_{\lambda_i})$ for all i , then A is **diagonalizable**!

After step 1, we can conclude that A is diag. or NOT

(Step 2) Finding invertible P and diagonal D .

- Invertible P

>>>> for each λ_i , we pick out $\alpha(\lambda_i) = \dim(E_{\lambda_i})$ linearly independent vectors from E_{λ_i} .

>>>> Therefore, we have $\sum_{i=1}^k \alpha(\lambda_i) = n$ vectors and they form the n columns of P .

- Diagonal D

>>>> For each eigenvector, place the corresponding eigenvalue on the diagonal.

if diag.

Diagonalization

Let A be an $n \times n$ -matrix.

(Step 1) Find all eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_s$.

- (Step 1a) If $\sum_{i=1}^k \alpha(\lambda_i) < n$, then A is **not** diagonalizable!

- (Step 1b) For each λ_i , check that $\alpha(\lambda_i) = \dim(E_{\lambda_i})$.

>>>> If $\alpha(\lambda_i) > \dim(E_{\lambda_i})$ for some i , then A is **not** diagonalizable!

>>>> If $\alpha(\lambda_i) = \dim(E_{\lambda_i})$ for all i , then A is **diagonalizable**!

(Step 2) Finding invertible P and diagonal D .

- Invertible P

>>>> for each λ_i , we pick out $\alpha(\lambda_i) = \dim(E_{\lambda_i})$

linearly independent vectors (**eigenvectors**) from E_{λ_i} .

>>>> Therefore, we have $\sum_{i=1}^k \alpha(\lambda_i) = n$ vectors and they form the n columns of P .

- Diagonal D

>>>> For each eigenvector, place the corresponding **eigenvalue** on the diagonal.

Example. Consider the following matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 0 & 3 \end{bmatrix}.$$

Recall that its characteristic polynomial is $(1 - \lambda)(2 - \lambda)(3 - \lambda)$.

Recall

$$\lambda_1 = 1 \Rightarrow E_1 = \left\{ s \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} : s \in \mathbb{R} \right\}$$

Find E_1

$$\lambda_2 = 2 \Rightarrow E_2 = \left\{ s \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} : s \in \mathbb{R} \right\}$$

Find E_2

$$\lambda_3 = 3 \Rightarrow E_3 = \left\{ s \begin{bmatrix} 5 \\ 8 \\ 2 \end{bmatrix} : s \in \mathbb{R} \right\}$$

Find E_3

$$\alpha(\lambda_i) = 1$$

(Step 1a) pass cause $\sum \alpha(\lambda_i) = 3$

$\alpha(\lambda_i)$

$\dim(E_{\lambda_i})$

1

1

1

1

1

1

$\therefore$ Step 1b pass!

Notation.

$\alpha(\lambda_i)$ = algebraic multiplicity of λ_i

$\dim(E_{\lambda_i})$ = geometric multiplicity of λ_i

$\therefore$ A diag. because $\alpha(\lambda_i) = \dim(E_{\lambda_i})$ for all λ_i .

Example. Consider the matrix

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Diagonalize A !

Diagonalizing A

(Step 1) Find all eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_s$.

- (Step 1a) If $\sum_{i=1}^k \alpha(\lambda_i) < n$, then A is **not** diagonalizable!

- (Step 1b) For each λ_i , check that $\alpha(\lambda_i) = \dim(E_{\lambda_i})$.

>>>> If $\alpha(\lambda_i) > \dim(E_{\lambda_i})$ for some i , then A is **not** diagonalizable!

>>>> If $\alpha(\lambda_i) = \dim(E_{\lambda_i})$ for all i , then A is **diagonalizable**!

What is algebraic multiplicity of $\lambda_1 = 0$?
i.e. what is $\alpha(\lambda_1)$?

char poly is $\det \begin{bmatrix} -\lambda & 1 \\ 0 & -\lambda \end{bmatrix} = \lambda^2$

eigen value $\lambda_1 = 0$

$$\therefore \alpha(\lambda_1) = 2$$

(Step 1a) pass! because $\sum \alpha(\lambda_i) = 2 = n$

Example. Consider the matrix

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Diagonalize A !

Diagonalizing A

(Step 1) Find all eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_s$.

- (Step 1a) If $\sum_{i=1}^k \alpha(\lambda_i) < n$, then A is **not** diagonalizable!

- (Step 1b) For each λ_i , check that $\alpha(\lambda_i) = \dim(E_{\lambda_i})$.

>>>> If $\alpha(\lambda_i) > \dim(E_{\lambda_i})$ for some i , then A is **not** diagonalizable!

>>>> If $\alpha(\lambda_i) = \dim(E_{\lambda_i})$ for all i , then A is **diagonalizable**!

What is geometric multiplicity of $\lambda_1 = 0$?
i.e. what is $\dim(E_{\lambda_1})$?

$$E_0 = \text{null}(A - 0 \cdot I)$$

$$\text{solve } \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$E_0 = \left\{ s \begin{bmatrix} 1 \\ 0 \end{bmatrix} : s \in \mathbb{R} \right\}$$

↓ vector

$$\therefore \dim(E_0) = 1$$

Example. Consider the matrix

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Diagonalize A !

Diagonalizing A

(Step 1) Find all eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_s$.

- (Step 1a) If $\sum_{i=1}^k \alpha(\lambda_i) < n$, then A is **not** diagonalizable!

- (Step 1b) For each λ_i , check that $\alpha(\lambda_i) = \dim(E_{\lambda_i})$.

>>>> If $\alpha(\lambda_i) > \dim(E_{\lambda_i})$ for some i , then A is **not** diagonalizable!

>>>> If $\alpha(\lambda_i) = \dim(E_{\lambda_i})$ for all i , then A is **diagonalizable**!

Is A diagonalizable?

1. Yes
2. ☒ No

Check that

$$\alpha(0) = 2 > \dim(E_0) = 1$$

$\therefore$ (step 1b) fails

$\therefore \hat{A}$ is not diagonalisable.

Example. Consider the matrix

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

Diagonalize A !

Diagonalizing A

(Step 1) Find all eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_s$.

- (Step 1a) If $\sum_{i=1}^k \alpha(\lambda_i) < n$, then A is **not** diagonalizable!

- (Step 1b) For each λ_i , check that $\alpha(\lambda_i) = \dim(E_{\lambda_i})$.

>>>> If $\alpha(\lambda_i) > \dim(E_{\lambda_i})$ for some i , then A is **not** diagonalizable!

>>>> If $\alpha(\lambda_i) = \dim(E_{\lambda_i})$ for all i , then A is **diagonalizable**!

What is the characteristic polynomial of A ?

characteristic polynomial is $\det \begin{bmatrix} -\lambda & -1 \\ 1 & -\lambda \end{bmatrix} = \lambda^2 + 1$

Here, $\lambda^2 + 1$ has no real roots

$\therefore$ sum of algebraic multiplicities is $0 < 2$

$\therefore$ (Step 1a) fails.

$\therefore \hat{A}$ is NOT diagonalisable

Example. Consider the matrix

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

Diagonalize A !

Here, $\hat{P} = \begin{bmatrix} -i & i \\ 1 & 1 \end{bmatrix}$

$$\hat{D} = \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix}$$

① $\hat{A}$ is NOT diag. using REAL matrices

BUT $\hat{A}$ is diag. using COMPLEX matrices

We only consider REAL matrices

TA; PI

diagonalize {{0,-1},{1,0}}

 NATURAL LANGUAGE

 MATH INPUT

 EXTENDED

Input interpretation

diagonalize

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

Result

$$M = S.J.S^{-1}$$

where

$$M = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$S = \begin{pmatrix} -i & i \\ 1 & 1 \end{pmatrix}$$

$$J = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}$$

$$S^{-1} = \begin{pmatrix} \frac{i}{2} & \frac{1}{2} \\ -\frac{i}{2} & \frac{1}{2} \end{pmatrix}$$

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When is A not diagonalizable

- Failure at **Step 1a**. That is, $\sum_{i=1}^k \alpha(\lambda_i) < n$.
→ this happens when char. poly does NOT have enough real roots
E.g. $1 + \lambda^2 \Rightarrow \sum \alpha(\lambda_i) = 0 < 2$
 $(1 + \lambda^2)(1 - \lambda)(2 - \lambda) \Rightarrow \sum \alpha(\lambda_i) = 2 < 4$
- Failure at **Step 1b**. That is, $\alpha(\lambda_i) > \dim(E_{\lambda_i})$ for some i .
This happens when E_{λ_i} does not have enough L.I. eigenvectors.

$$\dim(E_{\lambda_i}) < \alpha(\lambda_i)$$

Proposition (Vectors from Distinct Eigenspaces are Linearly Independent).

Let A be a real $n \times n$ -matrix and let $\lambda_1, \lambda_2, \dots, \lambda_s$ be distinct eigenvalues of A .
If v_i belongs to E_{λ_i} with $v_i \neq \mathbf{0}$, then $\{v_1, v_2, \dots, v_s\}$ is linearly independent.

*Proof in
colab

Proposition (geometric is at most algebraic).

Let A be a real $n \times n$ -matrix and let λ_i be an eigenvalue of A .

Then

$$1 \leq \overset{\text{geometric}}{\dim(E_{\lambda_i})} \leq \overset{\text{algebraic}}{\alpha(\lambda_i)}.$$

*Proof
omitted

 Proposition (Simple Spectrum implies Diagonalizable).

Let A be a real $n \times n$ -matrix.

If A has n distinct eigenvalues, then A is diagonalizable.

→ can immediately conclude

*Proof on
colab.

Why Diagonalize?

Today, we focus
→ combinatorics
→ population movement

Let Us Iterate Again

- Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$.
- Fix starting point $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} 1 \\ 1 \end{bmatrix} \mapsto \begin{bmatrix} 2 \\ 1 \end{bmatrix} \mapsto \begin{bmatrix} 3 \\ 2 \end{bmatrix} \mapsto \begin{bmatrix} 5 \\ 3 \end{bmatrix} \mapsto \begin{bmatrix} 8 \\ 5 \end{bmatrix}$$

These are Fibonacci numbers.

$$\begin{bmatrix} 21 \\ 13 \end{bmatrix} \leftarrow \begin{bmatrix} 13 \\ 8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

Example. Consider the following sequence

$$F_0 = 0, \quad F_1 = 1, \quad F_{k+2} = F_k + F_{k+1} \text{ for } k \geq 0.$$

Further, set $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$, $u_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $u_{k+1} = Au_k$ for $k \geq 1$.

What is u_{k+1} ? $= \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix}$

$$u_1 = \begin{bmatrix} F_1 \\ F_0 \end{bmatrix}$$

$$\tilde{u}_2 = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} F_1 \\ F_0 \end{bmatrix} = \begin{bmatrix} F_1 + F_0 \\ F_1 \end{bmatrix} = \begin{bmatrix} F_2 \\ F_1 \end{bmatrix}$$

Suppose that $\tilde{u}_k = \begin{bmatrix} F_k \\ F_{k-1} \end{bmatrix}$

$$\tilde{u}_{k+1} = \hat{A} \tilde{u}_k = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} F_k \\ F_{k-1} \end{bmatrix} = \begin{bmatrix} F_k + F_{k-1} \\ F_k \end{bmatrix} = \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix}$$

$$\begin{aligned} \tilde{u}_{k+1} &= \hat{A} \tilde{u}_k \\ &= \hat{A} \hat{A} \tilde{u}_{k-1} \\ &= \hat{A} \hat{A} \tilde{u}_{k-2} = \dots = \hat{A}^k u_1 = \hat{A}^k \begin{bmatrix} 1 \\ 0 \end{bmatrix} \end{aligned}$$

Example. Consider the following sequence

$$F_0 = 0, \quad F_1 = 1, \quad F_{k+2} = F_k + F_{k+1} \text{ for } k \geq 0.$$

Further, set $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$, $\mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{u}_{k+1} = A\mathbf{u}_k$ for $k \geq 1$.

$$\text{So, } \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix} = A^k \mathbf{u}_1.$$

Yes, if we are able to diagonalise $\hat{A}$,

$$\text{Suppose } \hat{A} = \hat{P} \hat{D} \hat{P}^{-1}$$

$$\hat{A}^k = (\cancel{\hat{P} \hat{D} \hat{P}^{-1}}) (\cancel{\hat{P} \hat{D} \hat{P}^{-1}}) \dots (\cancel{\hat{P} \hat{D} \hat{P}^{-1}})$$

$$= \hat{P} \hat{D}^k \hat{P}^{-1}$$

$$= \hat{P} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}^k \hat{P}^{-1}$$

$$= \hat{P} \begin{bmatrix} \lambda_1^k & 0 \\ 0 & \lambda_2^k \end{bmatrix} \hat{P}^{-1}$$

Example. Consider the matrix

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}.$$

Diagonalize A !

Diagonalizing A

(Step 1) Find all eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_s$.

- (Step 1a) If $\sum_{i=1}^k \alpha(\lambda_i) < n$, then A is **not** diagonalizable!

- (Step 1b) For each λ_i , check that $\alpha(\lambda_i) = \dim(E_{\lambda_i})$.

>>>> If $\alpha(\lambda_i) > \dim(E_{\lambda_i})$ for some i , then A is **not** diagonalizable!

>>>> If $\alpha(\lambda_i) = \dim(E_{\lambda_i})$ for all i , then A is **diagonalizable**!

What is the characteristic polynomial of A ?

1. $-1 - \lambda + \lambda^2$
2. Wrong answer

$$\det \begin{bmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{bmatrix} = -\lambda(1-\lambda) - 1$$

$$= -1 - \lambda + \lambda^2$$

$$= \left(\frac{1+\sqrt{5}}{2} - \lambda \right) \left(\frac{1-\sqrt{5}}{2} - \lambda \right)$$

$$= (\phi - \lambda) (\theta - \lambda)$$

Some rules

$$1 - \theta = \phi$$

$$\phi \theta = -1$$

golden ratio

phi $\leftarrow \phi = \frac{1+\sqrt{5}}{2} = 1.618..$

we drew xi $\leftarrow \theta = \frac{1-\sqrt{5}}{2} = -0.618..$
but idk how to draw

Example. Consider the matrix

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}.$$

Diagonalize A !

Corollary. Let A be a real $n \times n$ -matrix.

If A has n distinct eigenvalues,

then A is diagonalizable.

Is A diagonalizable?

$\hat{A}$ has two distinct eigenvalues
 $\Rightarrow \hat{A}$ is diagonalisable

$\lambda_1 = \phi \rightarrow$ eigenvector is $\begin{bmatrix} \phi \\ 1 \end{bmatrix}$
 $\lambda_2 = \theta \rightarrow$ eigenvector is $\begin{bmatrix} \theta \\ 1 \end{bmatrix}$ } after some work

$$\hat{P} = \begin{bmatrix} \phi & \theta \\ 1 & 1 \end{bmatrix}$$

$$\hat{D} = \begin{bmatrix} \phi & 0 \\ 0 & \theta \end{bmatrix}$$

Example. Consider the following sequence

$$F_0 = 0, \quad F_1 = 1, \quad F_{k+2} = F_k + F_{k+1} \text{ for } k \geq 0.$$

Further, set $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$, $\mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{u}_{k+1} = A\mathbf{u}_k$ for $k \geq 1$.

So, $\begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix} = A^k \mathbf{u}_1$. Moreover, $A = \begin{bmatrix} \phi & \xi \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \phi & 0 \\ 0 & \xi \end{bmatrix} \begin{bmatrix} \phi & \xi \\ 1 & 1 \end{bmatrix}^{-1}$.

$$\phi = \frac{1 + \sqrt{5}}{2} \approx 1.618 \dots$$

$$\xi = \frac{1 - \sqrt{5}}{2} \approx -0.618 \dots$$

$$\hat{A}^k = \begin{bmatrix} \phi & \xi \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \phi^k & 0 \\ 0 & \xi^k \end{bmatrix} \begin{bmatrix} \phi & \xi \\ 1 & 1 \end{bmatrix}^{-1}$$

$$= \frac{1}{\sqrt{5}} \begin{bmatrix} \phi & \xi \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \phi^k & 0 \\ 0 & \xi^k \end{bmatrix} \begin{bmatrix} 1 & -\xi \\ -1 & \phi \end{bmatrix}$$

$$= \frac{1}{\sqrt{5}} \left(\phi^k \begin{bmatrix} \phi \\ 1 \end{bmatrix} \begin{bmatrix} 1 & -\xi \end{bmatrix} + \xi^k \begin{bmatrix} \xi \\ 1 \end{bmatrix} \begin{bmatrix} -1 & \phi \end{bmatrix} \right)$$

$$= \frac{1}{\sqrt{5}} \left(\phi^k \begin{bmatrix} \phi & 1-\xi \\ 1 & -\xi \end{bmatrix} + \xi^k \begin{bmatrix} -\xi & \phi \\ -1 & \phi \end{bmatrix} \right)$$

g n (qy)

$$\hat{A}^k = \begin{bmatrix} \phi & \xi \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \phi^k & 0 \\ 0 & \xi^k \end{bmatrix} \begin{bmatrix} \phi & \xi \\ 1 & 1 \end{bmatrix}^{-1}$$

inverting matrix

$$= \frac{1}{\sqrt{5}} \begin{bmatrix} \phi & \xi \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \phi^k & 0 \\ 0 & \xi^k \end{bmatrix} \begin{bmatrix} 1 & -\xi \\ -1 & \phi \end{bmatrix}$$

$$= \frac{1}{\sqrt{5}} \left(\phi^k \begin{bmatrix} \phi \\ 1 \end{bmatrix} \begin{bmatrix} 1 & -\xi \end{bmatrix} + \xi^k \begin{bmatrix} \xi \\ 1 \end{bmatrix} \begin{bmatrix} -1 & \phi \end{bmatrix} \right)$$

$$= \frac{1}{\sqrt{5}} \left(\phi^k \begin{bmatrix} \phi & 1 \\ 1 & -\xi \end{bmatrix} + \xi^k \begin{bmatrix} -\xi & -1 \\ -1 & \phi \end{bmatrix} \right)$$

$$\phi = \frac{1 + \sqrt{5}}{2} \approx 1.618 \dots$$

$$\xi = \frac{1 - \sqrt{5}}{2} \approx -0.618 \dots$$

↳ this aka rank-one decomposition

↓ Too abstract, ignore

Theorem (Two-Term Linear Recurrences).

Let a_k be a sequence of numbers defined as follows.

$$a_0 = a_0, \quad a_1 = a_1, \quad a_{k+2} = c_0 a_k + c_1 a_{k+1} \text{ for } k \geq 0.$$

Let λ_1 and λ_2 be the roots of the characteristic equation $c_0 + c_1 \lambda - \lambda^2 = 0$.

Then there are constants B_1 and B_2 such that

$$a_k = B_1 \lambda_1^k + B_2 \lambda_2^k \text{ for all } k \geq 0.$$

Proof using diagonalisation

→ out of syllabus

Proof: TA; PI

Proposition (Koch Snowflake Sequence).

Let K_k be a sequence of numbers defined as follows.

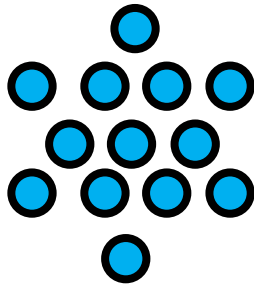
$$K_0 = 0, \quad K_1 = 1, \quad K_{k+2} = 13K_{k+1} - 36K_k \text{ for } k \geq 0.$$

Show that

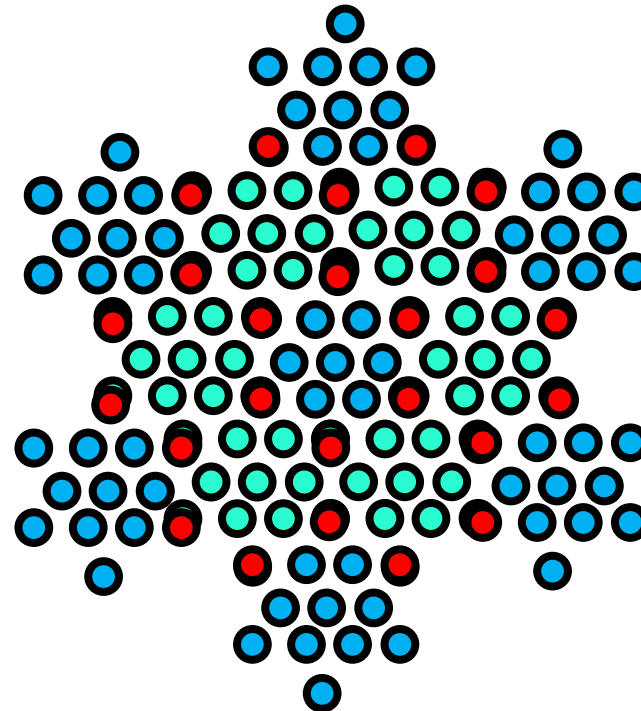
$$K_k = \frac{1}{5}(9^k - 4^k) \text{ for all } k \geq 0.$$



$$K_1 = 1$$



$$K_2 = 13$$

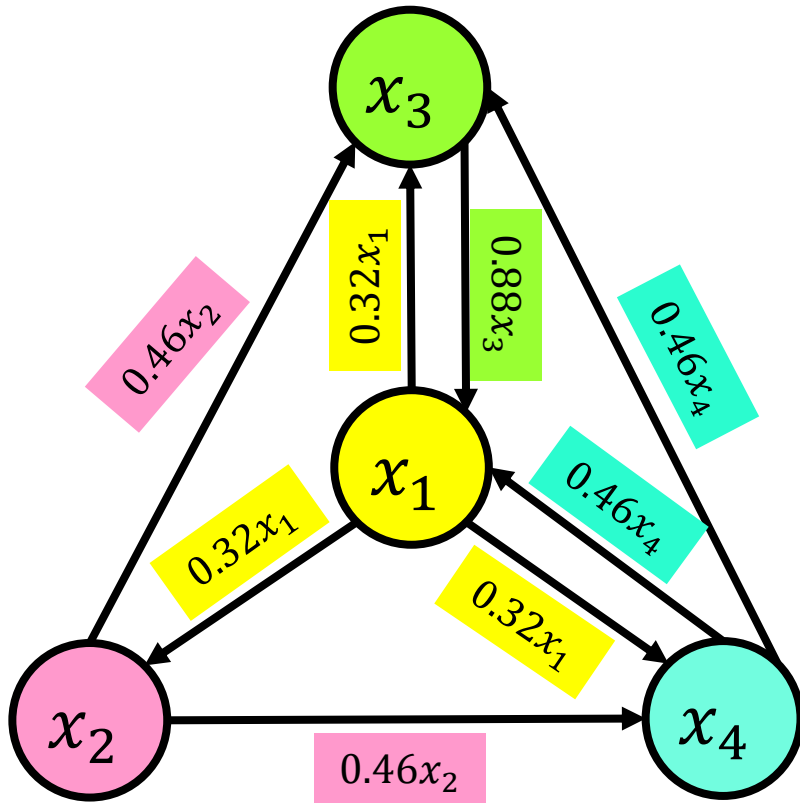


$$K_3 = 133 = 13 \cdot 13 - 36 \cdot 1$$

Tut 8, Q8
- Go through
on Monday

Population Movement

Consider four villages in country X. Each year, everyone move to another village according to the following rule.



This year x_1 x_2 x_3 x_4

	From Village 1	From Village 2	From Village 3	From Village 4
To Village 1	0.04	0.04	0.88	0.46
To Village 2	0.32	0.04	0.04	0.04
To Village 3	0.32	0.46	0.04	0.46
To Village 4	0.32	0.46	0.04	0.04

go copy

Simple Implementation

<https://editor.p5js.org/kiahhanmao/sketches/jPHOkIEAI>

Population Movement

When the population stabilizes, we have that

$$Mx = x.$$

g' copy

Theorem.

Let $\mathbf{M}$ be a square matrix whose columns sum to one.

Then 1 is an eigenvalue of $\mathbf{M}$. In other words, $\dim(E_1) \geq 1$.

Furthermore, if all entries of $\mathbf{M}$ are strictly positive, then the eigenspace corresponding to eigenvalue one has dimension exactly 1.

In other words, $\dim(E_1) = 1$.

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The \$25,000,000,000 Eigenvector: The Linear Algebra behind

HIDDEN

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success derives in large part from its PageRank algorithm, which ranks the importance of web pages according to an eigenvector of a weighted link matrix. Analysis of the PageRank formula provides a wonderful applied topic for a linear algebra course. Instructors may assign this article as a project to more advanced students or spend one or two lectures presenting the material with assigned homework from the exercises. This material also complements the discussion of Markov chains in matrix algebra. Maple and *Mathematica* files supporting this material can be found at www.rose-hulman.edu/~bryan.

Key words. linear algebra, PageRank, eigenvector, stochastic matrix